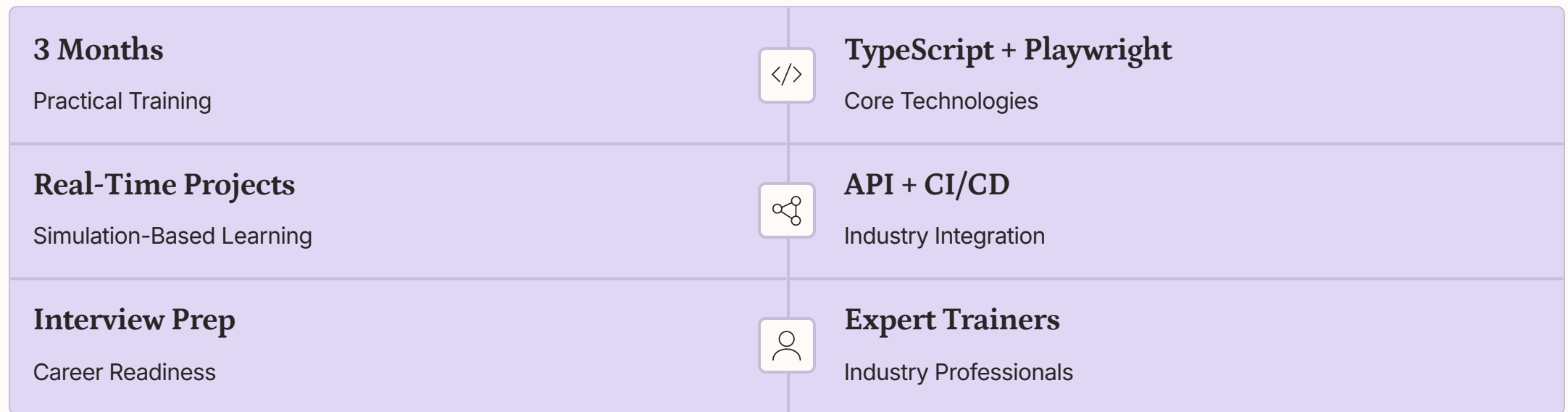


TypeScript & Playwright Automation

Master Automation Testing with Real-Time Projects

A comprehensive 3-month, hands-on training program designed for students, freshers, and job seekers who want to launch a high-impact career in Software Testing and Automation Engineering.



 **Join Our Next Batch — Limited Seats Available**

About XpertsEdgeTechnologies

XpertsEdgeTechnologies is a career-focused IT training institute dedicated to delivering practical, job-oriented training for students, freshers, and working professionals. Our curriculum is built around real-time examples, hands-on practice, and project-based learning that mirrors actual workplace environments. We go beyond theory — every session is designed to build confidence, sharpen technical skills, and prepare learners for real job interviews. With personalized trainer guidance, individual doubt support, and structured assessment cycles, we ensure every student is job-ready by the time they complete the program.

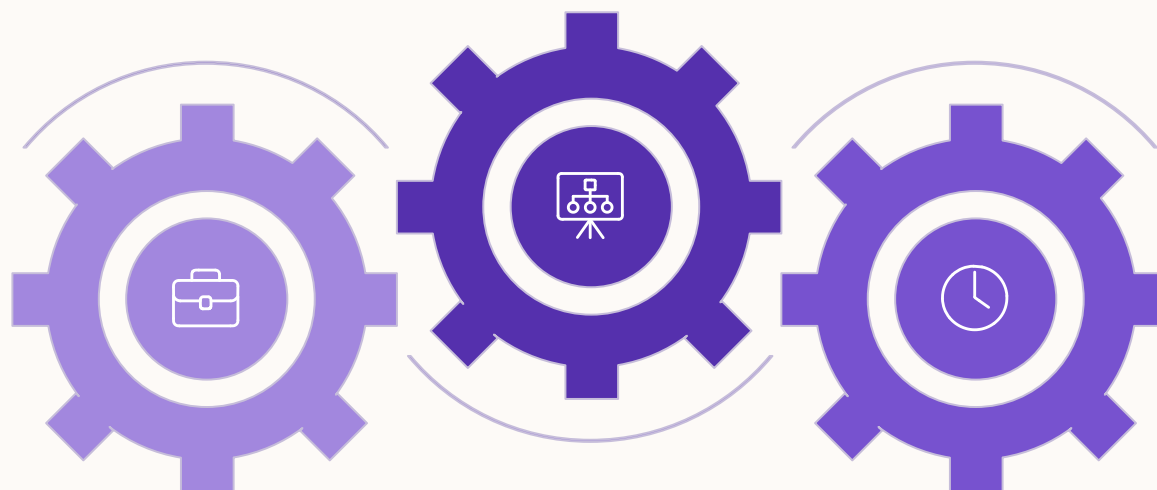
✔️ 🎓 Certificate of Completion awarded to all successful students upon finishing the program.

Why Choose Us?

- Experienced Industry Trainers
- Practical Hands-on Sessions
- Real-Time Automation Projects
- Beginner-Friendly Teaching Style
- Individual Doubt Support
- Job-Oriented Syllabus
- Mock Interview Preparation
- Certificate After Completion

Hands-On Projects

Apply skills with practical, real-world assignments.



Job-Oriented Training

Focused curriculum designed for career readiness.

Personalized Coaching

Receive one-on-one guidance from experts.

What You Will Learn in 3 Months

This intensive 3-month **TypeScript & Playwright Automation** course is structured to take you from zero to job-ready. Starting with TypeScript programming fundamentals and progressing through Playwright framework setup, UI automation, API testing, CI/CD integration, and real-time project development — every week builds on the last. The curriculum is designed to closely replicate the tools and workflows used in actual QA engineering roles.



Month 1

TypeScript Fundamentals & Automation
Basics



Month 2

Playwright Core Concepts & UI Automation



Month 3

Framework Design, CI/CD & Real-Time
Project



Learning Outcome: By the end of this course, students will be able to independently design, build, and execute a complete automation testing framework using TypeScript and Playwright — ready for real job environments.

Automation Basics + TypeScript Fundamentals

The first month establishes a strong technical foundation. Students are introduced to automation testing concepts, the Playwright ecosystem, and the TypeScript language — building the skills needed to write clean, reusable, and type-safe test code.

1

Intro to Automation & Setup

- Introduction to Automation Testing
- Why Playwright for Modern Testing?
- Role of TypeScript in Test Automation

2

TypeScript Basics

- TypeScript vs JavaScript
- Setting up tsconfig.json
- Variables, Data Types, Type Inference
- Functions and Arrow Functions

1

Object-Oriented Concepts

- Interfaces, Classes & Objects
- Inheritance, Abstraction, Polymorphism
- Access Modifiers, Abstract & Static Members

2

Advanced TypeScript

- Generics, Modules & Imports
- Promises and async/await
- Enums, Union & Intersection Types
- Utility Types: Partial, Pick, Readonly

🟢  **Outcome:** Students will master TypeScript fundamentals to write clean, reusable, and type-safe automation test code with confidence.

Playwright Core Concepts & UI Automation

Month 2 dives deep into the Playwright framework. Students learn to set up the test environment, locate UI elements precisely, interact with web applications, and validate behavior using Playwright's powerful assertion engine.



Playwright Fundamentals

- Architecture & Supported Browsers
- Installation & Configuration
- Test Runner Basics
- Launching Browsers & Contexts



Locators & Selectors

- CSS, Text & XPath Locators
- Locator Chaining & Filtering
- nth, hasText, has Strategies
- Locator Best Practices



Interactions & Validations

- Click, Fill, Select, Hover, Check
- File Uploads & Downloads
- expect() API & Assertions
- Visual Comparisons

📘 **✓ Outcome:** Students will identify UI elements, interact with web pages, and validate real-world application behavior using Playwright's full toolset.

Advanced Playwright: Waits, Alerts & Network

The second half of Month 2 covers advanced web automation challenges — dynamic element waits, dialog handling, multi-tab scenarios, frames, and network-level API interception. These are skills that differentiate a junior tester from a mid-level automation engineer.

Waits, Events & Alerts

Auto Waits & Manual Waits

`waitForSelector()` &
`waitForLoadState()`

Event Listeners & Alert Handling

File Chooser Events & Modal
Windows

Advanced Browser Handling

Multiple Tabs & Pop-ups

Frames & New Page Events

Network Activities

- Intercepting API Calls
- Mocking & Modifying Responses
- Validating Request & Response
- Monitoring Network Events

Network testing skills allow automation engineers to simulate backend failures, test loading states, and validate API contracts — all without leaving the browser.

 Mastering network-level testing is a key differentiator in senior QA roles.

Framework Design: Test Runner, Fixtures & POM

Month 3 elevates students from writing individual tests to designing scalable, maintainable automation frameworks. The focus shifts to architectural best practices — using Playwright's test runner configuration, custom fixtures, and the industry-standard Page Object Model (POM) design pattern.

1

Playwright Test Runner

- Test Configuration Files
- Parallel Execution Setup
- Projects and Devices Matrix
- Before/After Hooks

2

Fixtures & Test Hooks

- Creating Custom Fixtures
- Setup and Teardown Logic
- Shared Browser Contexts

3

Page Object Model

- What is POM and Why It Matters
- Creating Reusable Page Classes
- Centralizing Locators & Methods
- Base Page & Singleton Pattern

✅ **Outcome:** Students will design a maintainable, scalable Playwright automation framework using POM and reusable test setup patterns used in real companies.

API Testing, Test Data & Professional Reporting

A complete automation engineer must handle more than UI testing. This section covers Playwright's built-in API testing capabilities, test data management strategies, environment configuration, and integrating industry-standard reporting tools for professional test suites.

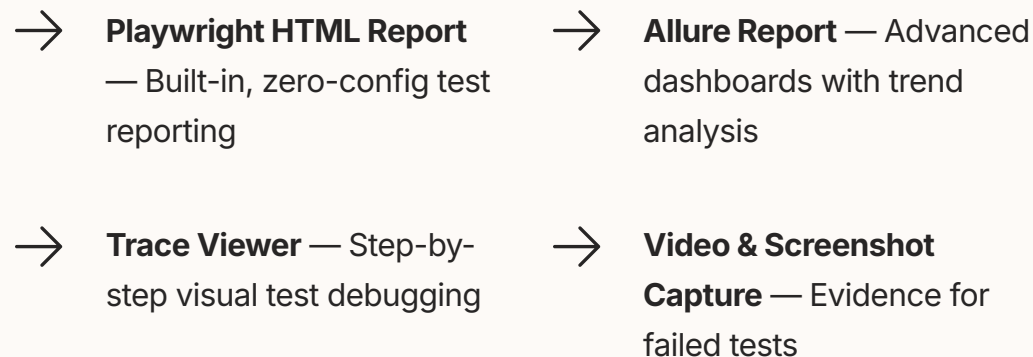
🔗 API Testing with Playwright

- Introduction to APIContext
- GET, POST, PUT, DELETE Methods
- Headers, Auth & Payloads
- Schema Validation
- Response Assertions

⚙️ Config & Data Management

- Environment Variables (.env)
- JSON & YAML Config Files
- Parameterization
- Data-Driven Tests

📊 Reporting Tools Covered



📌 **✓ Outcome:** Students will combine UI automation, API testing, data handling, and professional reporting into one complete suite.

CI/CD Integration & Real-Time Project Simulation

The final module brings everything together. Students integrate their automation suite into a CI/CD pipeline, apply framework best practices, and build a complete real-time automation project — simulating how QA teams operate in actual software companies.

CI/CD Integration

- GitHub Actions Pipeline Integration
- Running Tests in Docker Containers
- Scheduling Jobs using cron

Framework Enhancements

- Reusable Helper Utilities
- Custom Logging Mechanisms
- Error Handling Strategies
- Retry Logic for Flaky Tests

Real-Time Project Simulation



Build a Complete Automation Suite

Develop end-to-end test coverage for a real web application from scratch



Integrate UI + API + Reports

Connect all layers of the stack into a single professional test suite



Parallel Cross-Browser Execution

Run tests simultaneously across Chromium, Firefox, and WebKit

✅ **Outcome:** Students gain hands-on project experience building a production-grade automation suite — identical to what companies expect from automation engineers.

Start Your Automation Testing Career Today

🎯 Interview & Assessment Preparation

- Playwright + TypeScript Interview Questions
- Scenario-Based Practice Sessions
- Mock Assessments & Project Review
- Job Readiness Evaluation

💼 Career Roles You Can Apply For

Manual Tester

QA Engineer

Automation Tester

Playwright Automation Engineer

SDET

🚀 Join Our Next Batch — Seats Filling Fast!

Build your automation testing career with XpertsEdge Technologies. Enroll in our 3-month TypeScript & Playwright Automation program and take the first step toward your dream QA role.

🛠️ Tools You Will Master



TypeScript & Playwright

Core automation stack



Git, GitHub & GitHub Actions

Version control & CI/CD



Allure & HTML Reports

Professional test reporting



Docker & VS Code

Modern dev environment

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